EBU6505 – Reasoning and Agents
Block 2, Day 1 – Bayesian Networks

Dr Riasat Islam

Queen Mary University of London

Semester 1, 2026–2027
Teaching Week 4

Refs: AIMA4e Ch.12–13 pp.385–460; Poole3e Ch.9 pp.375–458

What have we covered so far this module?

- reasoning techniques for LLMs and test-time compute
- search and state-space problem solving
- MDPs and reinforcement learning
- knowledge representation and knowledge graphs
- graph-based reasoning for retrieval, memory, and agents

Main idea: Block 2 starts a new question: what should an agent believe when the world is uncertain and the evidence is incomplete?

What are we going to cover this week?

- Day 1: Bayesian networks
- Day 2: exact inference in Bayesian networks
- Day 3: probabilistic reasoning over time
- Day 4: planning with uncertainty
- Day 5: decision networks and POMDP synthesis

Main idea: This week moves from structured knowledge to uncertain knowledge: first model it, then infer with it, then plan with it.

Refs: AIMA4e Ch.12–13 pp.385–460; Poole3e Ch.9 pp.375–458

What are we going to cover today?

- why Bayesian networks are useful for uncertain reasoning
- what nodes, arrows, and CPTs mean
- how a network represents a full joint distribution
- how conditional independence and d-separation work
- what kind of inference question we will solve next

Main idea: Today is the foundation lecture. Students do not need heavy calculations yet; they need a clear mental model first.

Resources

- *Artificial Intelligence: Foundations of Computational Agents*, David Poole and Alan Mackworth (3rd ed.)
- Poole 3e, Chapter 9: Reasoning with Uncertainty
- *Artificial Intelligence: A Modern Approach*, Russell and Norvig (4th ed.)
- AIMA 4e, Chapter 12: Quantifying Uncertainty
- AIMA 4e, Chapter 13: Probabilistic Reasoning

Main idea: The two textbooks say the same core thing in different ways: a Bayesian network gives a graph structure plus probability tables.

Learning Outcomes for Today

- explain what a Bayesian network is in simple words
- read the meaning of nodes, arrows, and CPTs
- explain why the graph reduces the size of the probability model
- identify the three common d-separation patterns
- interpret a simple inference query with evidence

Main idea: If students leave with these five ideas clear, Day 2 calculations become much easier.

Warm-Up Question

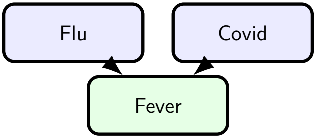
Question: If a patient has a fever, can we be 100% sure they have flu?

What we know

Fever is evidence, not proof.

Why

Different causes can lead to the same symptom.



Main idea: Uncertain symptoms need probabilistic reasoning, not yes-or-no rules.

What Problem Does a Bayesian Network Solve?

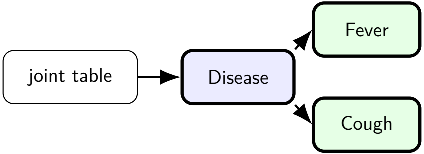
Question: Why not just write one huge probability table for every variable together?

One huge table

Hard to read, hard to estimate, and grows fast as we add variables.

Bayesian network

Split the model into small local pieces connected by a graph.



Main idea: A BN replaces one huge table with small connected pieces.

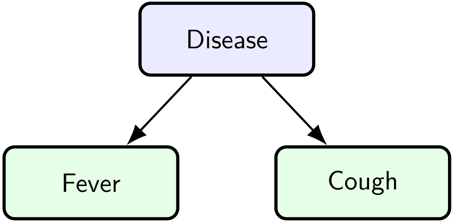
Refs: AIMA4e Ch.12–13 pp.385–460; Poole3e Ch.9 pp.375–458

What Is a Bayesian Network?

Question: What are the two parts of a Bayesian network?

- a **graph** that shows which variables directly depend on which others
- a **CPT** for each node that gives the probability numbers
- the graph is directed, and it must not contain a cycle

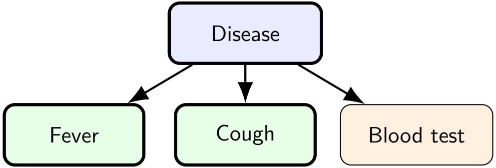
Main idea: A Bayesian network is a directed acyclic graph plus one local probability table for each variable.



$P(\text{Fever} \mid \text{Disease})$

How Do We Read a Small Bayesian Network?

Question: What does the graph say in simple words?



Node

one uncertain variable

Arrow

direct influence

CPT

local probabilities

Main idea: This graph says Disease directly changes the probability of Fever, Cough, and the Test result.

Why Is the Graph Better Than One Giant Table?

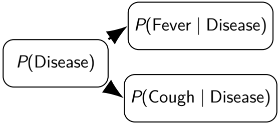
Question: What does the graph save us from writing down?

Without structure

- one big joint table
- many combinations
- hard to explain

With structure

- one small table per node
- local dependencies only
- easier to read and update



Main idea: The graph tells us which local tables we need.

Where Do CPT Numbers Come From?

Question: How do we fill in the probabilities on the nodes?

Experts

doctors, engineers, or domain specialists estimate the numbers

Data

count frequencies from records

Toy models

use temporary values for teaching or early design

Main idea: CPTs are not magic. They come from people, data, or temporary assumptions used for teaching and early design.

Example CPT: Fever Given Disease

Question: How do we read one local probability table?

Disease	$P(Fever = T)$	$P(Fever = F)$
Flu	0.8	0.2
No flu	0.2	0.8

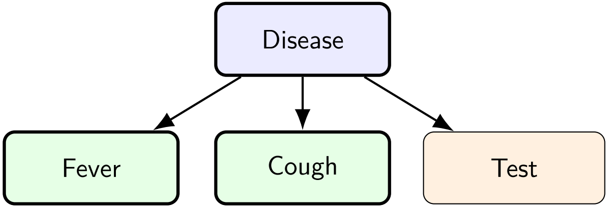
How to read it

If the patient has flu, the model gives a high probability of fever. If the patient does not have flu, fever is still possible, but less likely.

Main idea: A CPT is always local: one node, its parents, and the probabilities for that small situation only.

How Does the Whole Network Define One Probability Model?

Question: How do the local tables combine into one joint distribution?

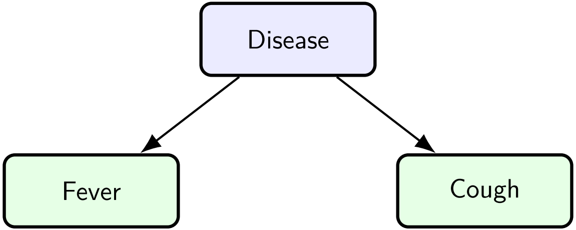


$$P(D, F, C, T) = P(D)\, P(F \mid D)\, P(C \mid D)\, P(T \mid D)$$

Main idea: The graph tells us how to break one large joint probability into smaller local factors.

What Does Conditional Independence Mean Here?

Question: When does one symptom stop giving extra information about another?



If Disease is already known, Fever tells us much less about Cough.

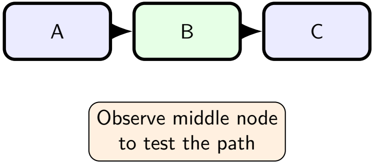
Main idea: Conditional independence means some variables stop sharing extra information once we already know the right middle variable.

Refs: AIMA4e Ch.12–13 pp.385–460; Poole3e Ch.9 pp.375–458

What Is D-Separation?

Question: How can we decide from the graph whether information can still flow along a path?

- pick a path between two variables
- look at the middle node pattern
- ask whether that middle node is observed
- decide whether the path is open or blocked

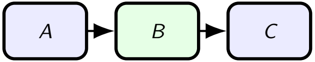


Main idea: Students can think of d-separation as an information-flow test: does evidence travel through this path or does the path stop it?

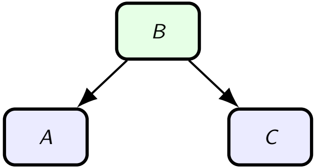
Refs: AIMA4e Ch.12–13 pp.385–460; Poole3e Ch.9 pp.375–458

Three Common D-Separation Patterns

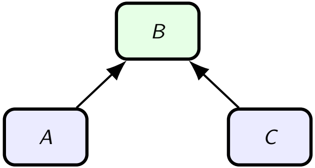
Question: Which middle-node pattern are we looking at?



Serial chain
observe B to block



Common cause
observe B to block



Collider
observe B to open

Main idea: The collider case is the unusual one. It behaves in the opposite way from the first two patterns.

Pattern 1: Serial Chain

Question: What happens in a chain when we observe the middle variable?

- structure: $A \to B \to C$
- before we know B, information can travel from A to C
- after we know B, the path is blocked

Example

Rain $\to$ Wet road $\to$ Traffic delay

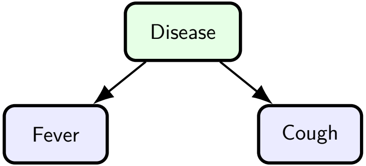
Main idea: In a serial chain, observing the middle node blocks the path.

Refs: AIMA4e Ch.12–13 pp.385–460; Poole3e Ch.9 pp.375–458

Pattern 2: Common Cause

Question: What happens when two effects share the same cause?

- structure: $A \leftarrow B \rightarrow C$
- before we know B, the two effects can look related
- after we know B, that path is blocked



Example

Fever $\leftarrow$ Disease $\rightarrow$ Cough

Main idea: In a common-cause pattern, observing the shared cause blocks the path between the effects.

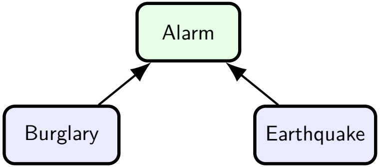
Pattern 3: Collider

Question: Why is the collider pattern different from the first two?

- structure: $A \to B \leftarrow C$
- before we know B, the path is blocked
- after we know B or its descendant, the path opens

Example

Burglary $\to$ Alarm $\leftarrow$ Earthquake



Main idea: In a collider, the path is closed by default. Observing the middle effect opens it.

Refs: AIMA4e Ch.12–13 pp.385–460; Poole3e Ch.9 pp.375–458

What Is Inference in a Bayesian Network?

Question: Once we have the network, what question do we usually ask?

Evidence we observe

Fever = true
Cough = true

Hidden variable we ask about

What is the probability of Disease?

$$P(\text{Disease} \mid \text{Fever} = T, \text{Cough} = T)$$

Read this as: probability of Disease given the evidence.

Main idea: Inference means updating belief about an unknown variable after we observe evidence.

Where Are Bayesian Networks Useful?

Question: What kinds of real tasks match this model well?

Healthcare

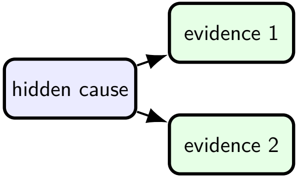
symptoms, tests, diseases

Systems

faults, alarms, missing sensors

Finance

market signals, fraud clues, loss risk



Main idea: Bayesian networks help when the real cause is hidden but we can observe uncertain clues.

Refs: AIMA4e Ch.12–13 pp.385–460; Poole3e Ch.9 pp.375–458

How Does Day 1 Connect to Day 2?

Question: What are we doing after we understand the model structure?

build the BN → add evidence → compute exact answers

Main idea: Day 1 is about building and reading the model. Day 2 is about computing answers from that model.

Key Takeaways

- A Bayesian network is a directed acyclic graph plus one CPT per node.
- The graph makes a large uncertain model smaller and easier to explain.
- Conditional independence is one of the main reasons the model stays compact.
- D-separation is the graph test for deciding whether a path is open or blocked.
- Inference asks for the probability of something unknown given the evidence we observe.

Main idea: The main lesson from Day 1 is simple: Bayesian networks separate structure from numbers, and that makes uncertain reasoning much easier to organise.

Refs: AIMA4e Ch.12–13 pp.385–460; Poole3e Ch.9 pp.375–458

In-Class Exercises

Think first → Work in pairs or small groups → Discuss and compare ideas

Use this time to work on today's in-class tasks and prepare one clear answer you can discuss.

Main idea: Use the exercise time to move from the picture of a Bayesian network to the meaning of the variables, edges, and query.

Further Study

- Read Poole Chapter 9 and AIMA Chapters 12–13.
- Practice: Poole Ch.9, Exercises 9.2, 9.3, 9.4, and 9.7.
- AIMA Bayesian-network exercise page: 14.1, 14.5, 14.6, and 14.8.
- Focus on building the graph, reading conditional dependence, and explaining what changes when evidence is observed.

Refs: AIMA4e Ch.12–13 pp.385–460; Poole3e Ch.9 pp.375–458